

Strategic and Predictive Analysis of Chess Openings

1. Motivation

Chess instruction commonly encourages amateur players to study the opening repertoires of elite grandmasters. It is not obvious, however, that lines which succeed at the top level transfer to club-level play, where games are decided far more often by tactical errors than by deep opening theory. This project asks a concrete question: **does a player's skill level meaningfully change which openings appear, and how predictable are game outcomes from rating and opening information?** If opening usage depends on rating, then effective preparation should be tailored to a player's actual strength rather than copied from professional theory. The goal is to test this empirically and to build a classifier that estimates the likely winner of a given matchup.

2. Data Source

The primary dataset is the **Chess Game Dataset (Lichess)**, obtained from Kaggle, containing roughly 20,000 online games. Relevant fields include `white_rating`, `black_rating`, `opening_name`, `turns`, and `winner`. After dropping records with missing ratings, opening, or result, **20,058 games** remained for analysis.

The dataset was complemented by a **data-enrichment pipeline built on the official Lichess API** using the `berserk` Python client. A token-authenticated session retrieves live game metadata directly from active player profiles (`client.games.export_by_player`), validating that the dataset can be refreshed or extended with current games rather than being limited to a fixed historical snapshot.

For analysis, the continuous Elo ratings were discretized into three skill tiers based on White's rating:

- **Beginner:** rating ≤ 1200
- **Intermediate:** $1200 < \text{rating} \leq 1800$
- **Advanced:** rating > 1800

3. Data Analysis

The analysis proceeded in four stages.

Stage 1 — Preprocessing and feature engineering. Rows with missing key fields were removed, ratings were binned into the three skill tiers above using `pandas.cut`, and the categorical fields (`opening_name`, `rating_bracket`) were label-encoded for modeling.

Stage 2 — Exploratory Data Analysis. Analysis was focused on the **10 most frequent openings** in the dataset. Three views were produced: a box plot of White rating by game result (outlier inspection), a correlation heatmap of `white_rating`, `black_rating`, and `turns` (rating vs. game length), and a normalized stacked bar chart of win/draw/loss rates across the top 10 openings.

Stage 3 — Inferential statistics. A **Chi-Square Test of Independence** (`scipy.stats.chi2_contingency`) was run on the contingency table of the top 10 openings against the three rating brackets, at significance level $\alpha = 0.05$, to test whether opening usage is statistically associated with skill tier.

Stage 4 — Predictive modeling. A **Random Forest Classifier** (`n_estimators=100`) was trained on `white_rating`, `black_rating`, the encoded `opening_name`, and the encoded `rating_bracket` to predict the `winner` (white / black / draw). The data was split 80% train / 20% test. Model quality was assessed with an accuracy score, a per-class classification report, a confusion matrix, and a feature-importance ranking. A helper function, `predict_matchup`, was implemented to run inference on individual player-versus-player scenarios and report a confidence value.

4. Findings

Opening usage depends significantly on skill level. The Chi-Square Test of Independence yielded $\chi^2 = 248.3951$ with $p = 1.7287 \times 10^{-42}$. Because the p-value is far below $\alpha = 0.05$, the null hypothesis of independence is rejected. This provides strong statistical evidence that the distribution of openings differs across rating tiers, supporting the view that an opening repertoire should be matched to a player's actual strength rather than imitated from grandmaster play.

The EDA shows a clear behavioral shift. Across the top 10 openings, lower-rated players concentrate on open, intuitive systems, while higher-rated players shift toward more complex, theory-heavy lines such as the Sicilian Defense. The correlation analysis also indicates only a weak relationship between rating and game length.

Outcomes are predictable but only moderately so. The Random Forest achieved a test **accuracy of 62.69%**, clearly above the no-information baseline but far from deterministic. Per-class results were uneven: White and Black were predicted reasonably ($f1 \approx 0.66$ and 0.62), whereas draws were predicted poorly ($f1 \approx 0.14$) because they are rare in the data. As an example, the `predict_matchup` engine forecasts a 1500-rated White player versus a 1400-rated Black player in the Sicilian Defense as a **predicted Black win with 60.00% confidence** — a modest figure that reflects the genuine variance of club-level games, where a small rating edge does not strongly determine the result.

5. Limitations and Future Work

Several limitations qualify these results:

- **Platform and format bias.** All games come from Lichess online play. Online rapid/blitz games differ from over-the-board and classical play, so the findings may not generalize to all competitive contexts.
- **Enrichment not yet scaled.** The Lichess API pipeline is implemented and validated, but the modeling still runs on the ~20,058 Kaggle records; the large-scale enrichment envisioned in the proposal has not been applied.
- **Tier thresholds are heuristic, and based on White only.** The 1200/1800 cutoffs are reasonable but arbitrary, and the bracket uses only White's rating, ignoring Black's strength when forming tiers.
- **Limited feature set.** The model uses ratings, opening, and tier only. It ignores time control, color-specific tendencies, individual style, and move-level play, all of which affect outcomes.
- **Weak on draws.** The classifier almost never predicts draws (recall ≈ 0.09) because of class imbalance, capping overall accuracy at ~63%.
- **Direction of effect untested.** The Chi-Square test establishes that opening distribution differs across tiers but does not formally quantify *which* openings drive the difference; the directional reading comes from the EDA charts rather than a statistical decomposition.

Future work could address these by: applying the Lichess API enrichment to substantially increase sample size and balance the rating tiers; adding time-control and move-level features (e.g., centipawn loss or opening accuracy); using a rating gap or both players' ratings to form tiers; addressing the draw class imbalance via resampling or class weighting; performing standardized residual analysis on the contingency table to identify which opening–tier cells contribute most to the χ^2 statistic; and benchmarking alternative models (gradient boosting, logistic regression) against the Random Forest.